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Different styles of remagnetization in Devonian sediments from the north-western Sahara (Algeria)

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SUMMARY

Sediments of Devonian age located in three areas of North Africa have been affected to different degrees by the Hercynian Orogeny. They show various remagnetizations of Carboniferous, Permian and Pleistocene age. In the Ben–Zireg anticline near the Variscan front in Algeria, the main phase of magnetic overprinting occurs during the Tournaisian. It is superimposed on a Famennian primary direction carried by magnetite. In the Béni–Abbès area to the South of the Variscan belt, griotte limestones of Famennian age show a strong recent remagnetization with few residual Permian directions. This overprinting shows a shallower inclination than the present-day field, with a small westward declination. Remagnetization is associated to magnetic transformation of sulphides. The transition of sulphides into magnetite-type minerals, as revealed in the thermomagnetic curves from rocks of the Variscan Belt, preserved the carrier against future magnetic overprinting. Compared to partially remagnetized rocks, this mechanism is referable to recrystallization during deformation.

Key words: Africa, palaeomagnetism, Palaeozoic, remagnetization, rock magnetism, Variscan belt.

1 INTRODUCTION

Different Palaeozoic palaeogeographic models, mainly based on palaeomagnetic data, are proposed for the drift history of Gondwana (Irving 1977; Morel & Irving 1978; Ziegler 1990; Bachtadse & Briden 1991). In these reconstructions the apparent polar wandering path (APWP) is also built from data collected in the Southern Hemisphere (South America and Australia). The lack of stability tests, such as the fold test (McElhinny 1964), when dating the acquisition of the natural remanent magnetization (NRM) are often the origin of conflicting interpretations. The multicomponent character of the magnetization of Lower Palaeozoic rocks collected in Africa reveals Mid to Late Palaeozoic overprints (Morris & Carmichael 1978; Kent, Dia & Sougy 1984; Perrin, Elston & Moussine-Pouchkine 1988) as well as primary directions of the NRM (Bachtadse, Van der Voo & Haelbich 1987). Demagnetization techniques must be carefully undertaken in order to isolate all magnetic components. Even if some recent progress in determining new palaeopoles from the Algerian Saharan craton (Aïfa, Feinberg & Pozzi 1990; Merabet *et al.* 1991; Henry *et al.* 1992) has been achieved, the African APWP is still poorly constrained for Palaeozoic times (Bachtadse & Briden 1989, 1990, 1991; Feinberg *et al.* 1990). This is especially true for the key period of the Devonian. Because of large time gaps in the APWP due in part to remagnetization problems which often occurred during the Permian, palaeomagnetic data might not be completely rejected if they can throw light on their mechanism of occurrence. Problems of reliability of data (McFadden & Jones 1981; McFadden & McElhinny 1990) still exist since no new methodology is available to use the large data base of overprinting. If the components of magnetization are well determined using least-squares techniques, their absolute age is not often constrained. In fact, the characteristic component of magnetization is related to the stability of the magnetic carrier, whose age can be either related to the age of deposition or to later events such as diagenesis or recrystallization.

It has been reported that slightly deformed rocks of the Saharan platform (Ougarta) have been completely remagnetized during one or several tectonic events (e.g. Permian and/or Pleistocene), whereas some deformed rocks from the Variscan chain (Béchar basin) have not been affected (Aïfa 1987; Aïfa *et al.* 1990). This observation is true for North African Palaeozoic formations (Aïfa 1987; Salmon 1988; Aïfa *et al.* 1990) but not for the Palaeozoic Appalachian (Stamatakos & Hirt 1992) rocks of continental origin. The study of the Famennian griotte limestones (Fig. 1) from two

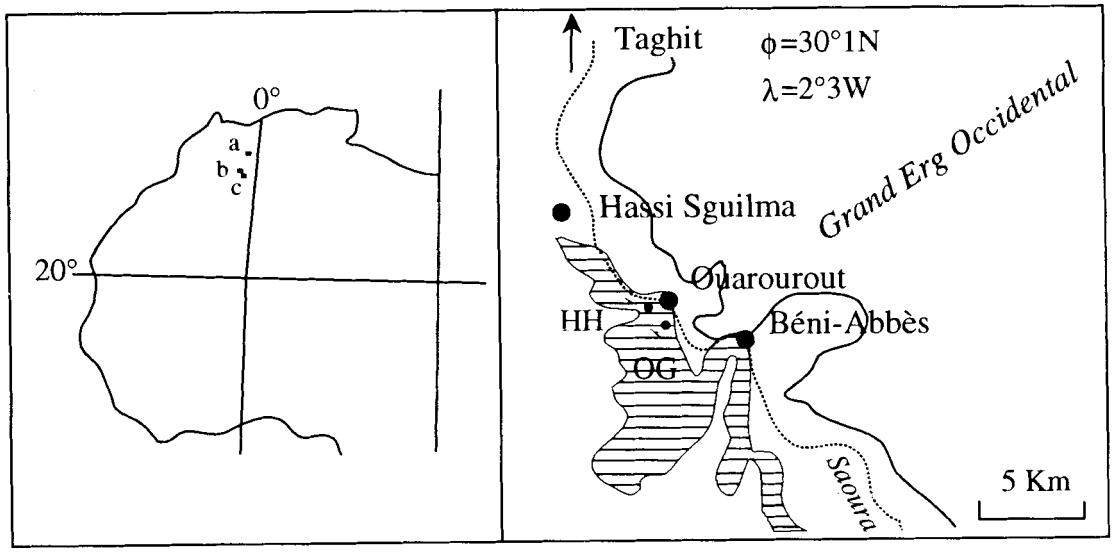


Figure 1. Location map and sampling sites in the north-western Sahara. The left half of this figure represents the Ben-Zireg (a), the Ouarourout (b) and the Béni-Abbès (c) geographic positions; the right half shows the sampling sites in the Ouarourout anticline, noticed HH and OG. Note that the series in (a) and (b) are Famennian griotte limestones whereas the formations in (c) are Lower Devonian Orthoceras rich limestones. The dashed area represents the Palaeozoic outcrops where Famennian griotte limestones have been cored. For detailed lithology see Pareyn (1961).

regions (North African Hercynian belt and Saharan platform) is here compared in order to provide a better understanding of the mechanism of remagnetization phenomena and to supplement existing palaeomagnetic data (Aïfa *et al.* 1990) for sites (a) and (b) (Fig. 1).

2 PALAEOMAGNETISM

Not far from the Variscan belt, primary or prefolding directions of magnetization have been isolated (Salmon *et al.* 1988; Aïfa *et al.* 1990) whereas, Permian and/or recent overprinting (Aïfa 1987; Merabet *et al.* 1991; Henry *et al.* 1992) have been described mainly from Saharan platform rocks. The Ouarourout Famennian griotte limestones are locally deformed by decametric folds forming an anticline which shows dips up to 50°. A large part of the sediments are probably alterated due to the oxidation environment close to the Saoura River (Fig. 1). Yet the samples have been cored in fresh rocks. The geological setting of this area has already been studied in detail by Pareyn (1961).

Ouarourout outcrop is practically of same age and lithology as the Ben-Zireg griotte limestones. It is intercalated between marls at the base of the typical sandstone facies of Ouarourout called the Saoura series (Pareyn 1961). The strata thickness does not exceed 20 m.

52 samples have been cored from Ouarourout (Fig. 1) over an area of 1 km^2 using a portable gasoline-powered drilling machine and orientated with both a magnetic or sun compass and an inclinometer. The 2.5 cm in diameter cores were cut in the laboratory into specimens of 2.5 cm length. Natural remanent magnetization (NRM) has been measured using a cryogenic LETI magnetometer with a noise level of $5 \cdot 10^{-6}$ A m^{-1}. The intensities of the NRM show a unimodal distribution with a maximum at around $27 \cdot 10^{-3}$ A m^{-1}. NRM directions clustered into two groups, one close to the present-day field and another close to the Permian direction as expected for this area (Fig. 2).

The specimens were cleaned by stepwise thermal demagnetization up to 600 °C or higher using a Schonstedt TSD-1 oven. The direction of remanent magnetization remaining after each demagnetization step was plotted on Zijderveld (1967) orthogonal projections and isolated by a least-squares method (Kirschvink 1980). AF demagnetization did not result in a better separation of components than thermal cleaning, thus the results obtained by this method will not be considered further in this paper.

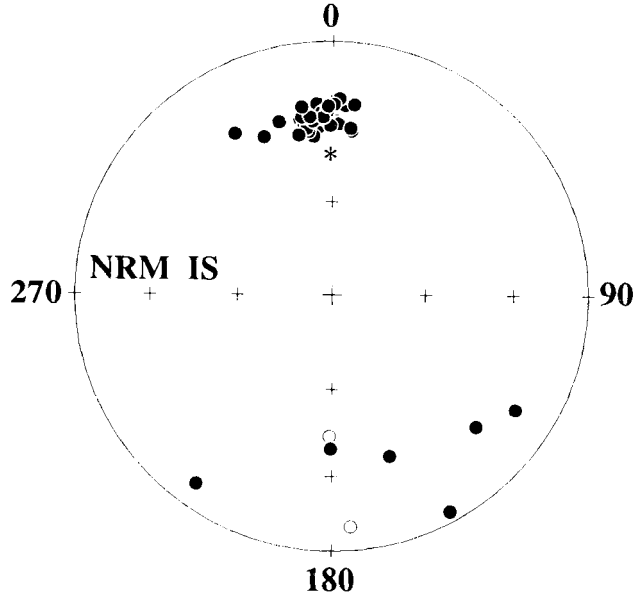


Figure 2. Stereonet projection diagram of the NRM directions for specimens from Ouarourout (sites HH and OG). Open (solid) circles indicate upper (lower) hemisphere. The star shows the present dipole field.

RESULTS

Magnetization components

Thermal demagnetization techniques reveal two components of magnetization (Figs 3a–f): a viscous component mainly close to the present dipole field, destroyed before 200 °C, and a high unblocking temperature component (up to 620 °C) labelled A1, A2 or B (Fig. 3i). Before tilt corrections, two characteristic directions are isolated, one close to the present dipole field with a low inclination (Figs 3h and 4a), whereas the other is close to the Permian direction as expected for this region, compared to the proximity of the Béni-Abbès sites (Aïfa *et al.* 1990). After

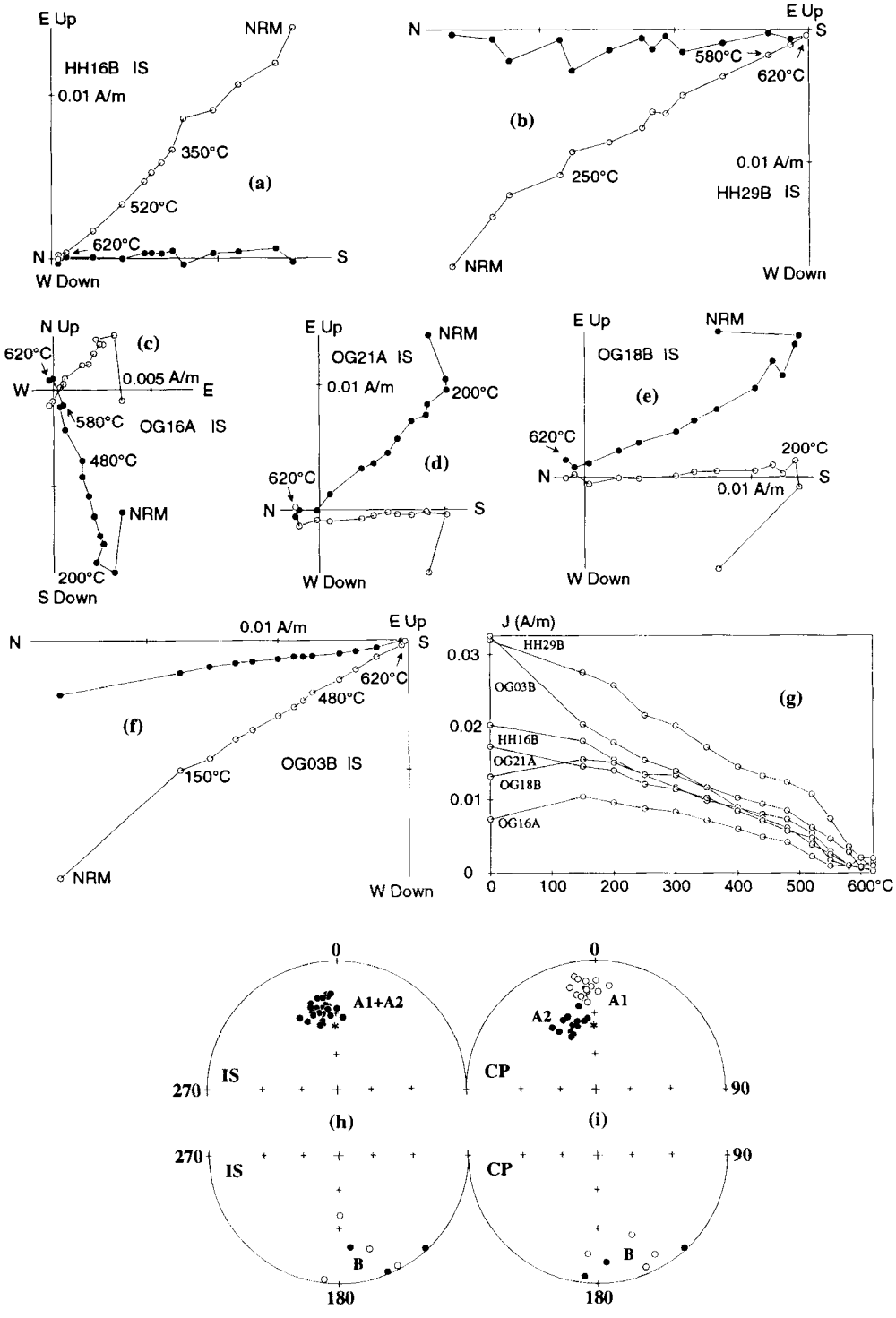


Figure 3. Orthogonal projection (Zijderveld 1967) of the thermal demagnetization for specimens from sites HH and OG (a–f) and their intensity decay (g) curves. Open (solid) circles indicate projection onto the vertical (horizontal) plane. Three main components named A1, A2 and B are isolated. Stereonet projection diagrams (h–i) of characteristic components of magnetization before (IS) and after (CP) tilt corrections, with a negative fold test (McElhinny 1964), showing two periods of overprints.

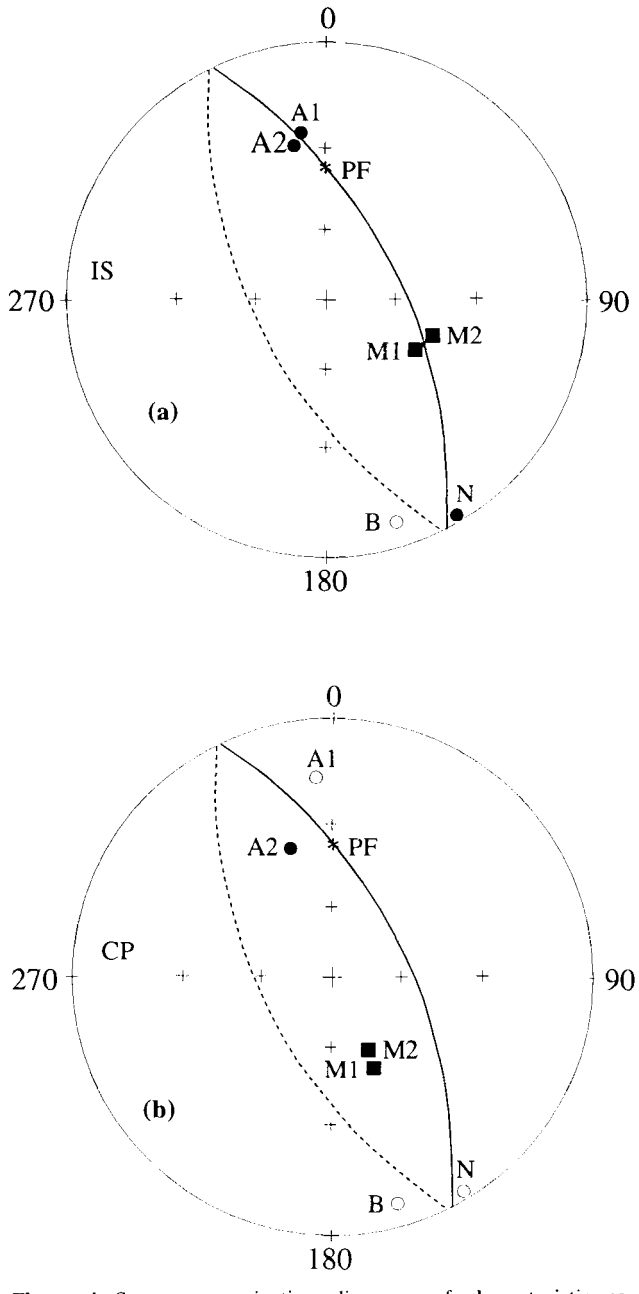


Figure 4. Stereonet projection diagrams of characteristic components of magnetization before (a) and after (b) tilt corrections. It shows two periods of overprints: Permian, Pleistocene with a slightly low inclination, along a great circle through the present dipole field PF (39°N, 4°30'W) and Permian directions B and N, before tilt corrections (IS). *M*2 and *M*1 are respectively Famennian and Tournaisian directions of magnetization for Ben-Zireg sites (see Table 1).

tilt corrections, a dispersion occurred (Table 1), the cluster separated into two groups of positive and negative inclinations (Figs 3i and 4b) but the Permian cluster did not vary significantly.

Rock magnetism

Rock magnetism experiments have been performed on Devonian rocks (specimens from sites BZ, BJ, BE, HH and OG) to determine the magnetic carriers. Fig. 5 represents thermomagnetic experiments using a Curie balance, under vacuum conditions [(V), reducing environment] and at atmospheric pressure [(A), oxidizing environment]. Enlargement diagrams corresponding to the heating part of the curves from the first column are shown in the second column. They reveal three different behaviours. (1) one principally during the heating phase of the experiment with frequent transformations around 500–550 °C (Figs 5a, b and e), (2) while the other with a loss of magnetization after 400–500 °C (Figs 5c and d), especially observed under vacuum conditions. (3) Meanwhile a different but a particular behaviour is observed for specimen OG07 (Fig. 5f).

Thermomagnetic curves suggest mineralogical changes between 500 °C and 550 °C, referable to sulphides (may be of pyrrhotite type) or titanium rich magnetite which transformed into magnetite. This seems to be characteristic of specimens from Ben-Zireg sites [site (a) of Fig. 1]. The large amount of magnetite produced is then referable to a large amount of sulphides under vacuum conditions (Figs 5a, b and e). Haematite is then produced at atmospheric pressure as shown in thermomagnetic curves of specimens BZ02(A), BJ15(A), OG07(A) (Fig. 5).

The cooling part of the curves defines only one Curie point around 580 °C, related to production of magnetite, which occurred with a large amount on specimens of behaviour (1), characterized by high NRM intensities ($21 \cdot 10^{-3}$—$33 \cdot 10^{-3}$ A m^{-1}). In specimens of behaviour (3), the main magnetic carrier is probably magnetite associated to sulphides as suggested by the two thermomagnetic experiments performed on sample OG07, respectively under vacuum conditions and at atmospheric pressure. In fact, above 400 °C, sulphides transformed into haematite which remains stable in an oxidizing environment [OG07(A)] or into magnetite under reducing conditions [OG07(V)]. Magnetic susceptibility, observed in curves of Fig. 6e, changes slightly between 400–500 °C (especially for specimen HH06B), but is significant after 550–600 °C. Meanwhile, specimens of Figs 6a and c show changes in the isothermal remanent magnetization (IRM) curves after heating beyond 500 °C. It means that some weak transformations occurred before 500 °C, which are also visible in the IRM demagnetization curves (Figs 6b and d). This is in agreement with the specimens of behaviour (2). We notice also that the reducing environment is linked to a reversible thermomagnetic curve (Fig. 5f) which shows two Curie points, one around 500 °C and the other at 580 °C, thus revealing the presence of two carriers referable to sulphides or likely magnetite, whereas the oxidizing environment is linked to an irreversible thermomagnetic curve [Fig. 5, specimen OG07 (A)] which does not show clearly the two Curie points mentioned but does show haematite.

IRM curves (Figs 6a–d) show mainly a high coercive mineral (probably goethite or haematite) at room temperature. At 500 °C, (Fig. 6c), a rapid increase in magnetization occurred at weak inducing fields, without noticeable variation in susceptibility [$2.156 \cdot 10^{-4}$ ($1.835 \cdot 10^{-4}$) before (after) heating]. This corroborates the change in mineralogy with fabric of a weak coercive mineral associated to probably magnetite. As goethite is destroyed

Table 1. Mean characteristic directions of sites from the studied area. *n* is the number of specimens or sites used to compute the mean vaulue. Db (Da), Ib (Ia), Kb (Ka) and $\alpha_{95}b$ ($\alpha_{95}a$) are respectively the values of declination, inclination and statistical parameters associated to the mean value (Fisher 1953) before (after) tilt corrections. A: Pleistocene; B,N: Permian; M2: Fammenian; M1: Tournaisian (see Aïfa *et al.* 1990 and Fig. 4)

Great circle analysis	Q	Dq (°)	Iq (°)
mean.is	0.000061	243.8	31.5
mean.cp	0.001653	251.2	1.1

Q, DQ, Iq are the parameters of great circles poles, mean.is (cp) are the mean values before (after) tilt corrections. Q is the precision parameter for great circle (see e.g. Halls 1978)

	Directions			Fishers's Parameters	
Components	Db (Da)	Ib (Ia)	n	kb (Ka)	α_{95}b(α_{95}a)
A1	352 (355)	24.6 (-16.1)	19	148	2.8
A2	347.9 (341.2)	28.7 (35.9)	14	108(91)	3.8 (4.2)
B	162.8 (164)	-5.5 (-5)	7	10 (14)	19.5 (16.7)
M2	109.6 (154.9)	44.1(54.3)	54	6 (24)	7.6 (3.7)
N	149.6 (149.3)	1.6 (-1.6)	52	8	6.8
M1	(156.4)	(47.6)	3	(8586)	(0.9)

before 200 °C, hence this mineral could be a sulphide of pyrrhotite type.

At room temperature, the IRM curve for specimen HH08A (Fig. 6a) displays saturation before 0.1 T, but does not show any saturation before 1.2 T (which is considered to be related to two carriers of different coercivities). This is probably explained by the presence of haematite and magnetite or titanium-rich magnetite. For specimen OG10A (Fig. 6c), before heating we may notice the presence of a single carrier which is probably of haematite type since no saturation occurred before 1.2 T in the IRM curve. After heating at 500 °C, the corresponding curve shows a behaviour similar to that observed for the previous case. This may be interpreted as an occurrence of a low percentage of magnetite in this sample. We mainly notice that the presence of magnetite is related to an irreversible thermomagnetic curve (Fig. 5f).

The examination of thin sections in reflected light microscopy (J. J. Chauvel) for each site also reveals this duality. Haematite and goethite are observed together in sites HH and OG, whereas haematite has been observed in sites BZ, BE and BJ. Nevertheless, secondary minerals such as pyrite were occasionally observed with some pyrrhotite inclusions.

All these observations indicate that the NRM is carried by a high coercivity mineral with maximum unblocking temperature below 600 °C. The intensity decay curves (Fig. 3g) corroborates this maximum unblocking temperature of 600 °C. Goethite is present as it breaks down at temperatures below 200 °C, characteristic magnetization is thus believed to be magnetite associated to a sulphide type. Sulphides are unstable below 500 °C (Fig. 6e) and are usually transformed into magnetite and/or haematite.

According to Scanning Electron Microscopy (SEM) observations of thin sections from specimens BJ and CB, we identified pyrite, magnetite and haematite (Fig. 7). The origin of inclusions of pyrrhotite in pyrite which transforms into magnetite between 300 °C and 400 °C is probably diagenetic due to pyrite precursor, and to a slow oxidation mechanism which may result in the formation of magnetite (Schwartz 1975; Freeman 1983, 1986). Such a phenomenon has been identified in calcareous rocks from Spitzbergen (Jelenska & Vincenz 1987). The oxidation conditions necessary to develop these transformations could be due to changes in the oxido-reduction of the environment under which these minerals were deposited. Two hypotheses are consequently possible. One suggests that after some tectonic deformation, the sediments were put in contact with the oxidizing atmosphere. The other envisages that fluid circulation was responsible for oxidation (during Permian times, see e.g. Oliver 1986).

3 DISCUSSION AND CONCLUSION

The remagnetization periods observed in the north-western Sahara may be correlated with the successive tectonic phases in this studied area (Fig. 8). *The Hercynian folding phase* (360–340 Ma) is well documented in the Ben–Zireg structure (Pareyn 1961) where rocks of the last strata of Upper Famennian are lacking and unconformity of Lower Carboniferous (Visean) age is evident. This phase probably preserved primary remanent magnetization, carried by magnetite, explained by a positive fold test (Table 1, component *M2* with $\alpha_{95} = 3.7°$). The Tournaisian–Visean phase instigated a partial remagnetization (Table 1, component *M1* with $\alpha_{95} = 0{\cdot}9°$) since carriers had undergone a 'vaccination' and became 'hard'. The effects of the first folding correspond with large wavelength deformations and tilting. These folds which indicate weak deformation in the Saharan platform are comparable to those of the Appalachan orogen generated in the same period of time. After the first Devonian–Carboniferous deformation phase, deformation lasted up to Permian times during a period of crustal shortening. On the contrary, except in the Reggane region, there is no unconformity in the Carboniferous strata which could indicate recent folding in the Ben–Zireg area. To the South, Middle Carboniferous folds have been evidenced (Conrad & Lemosquet 1984).

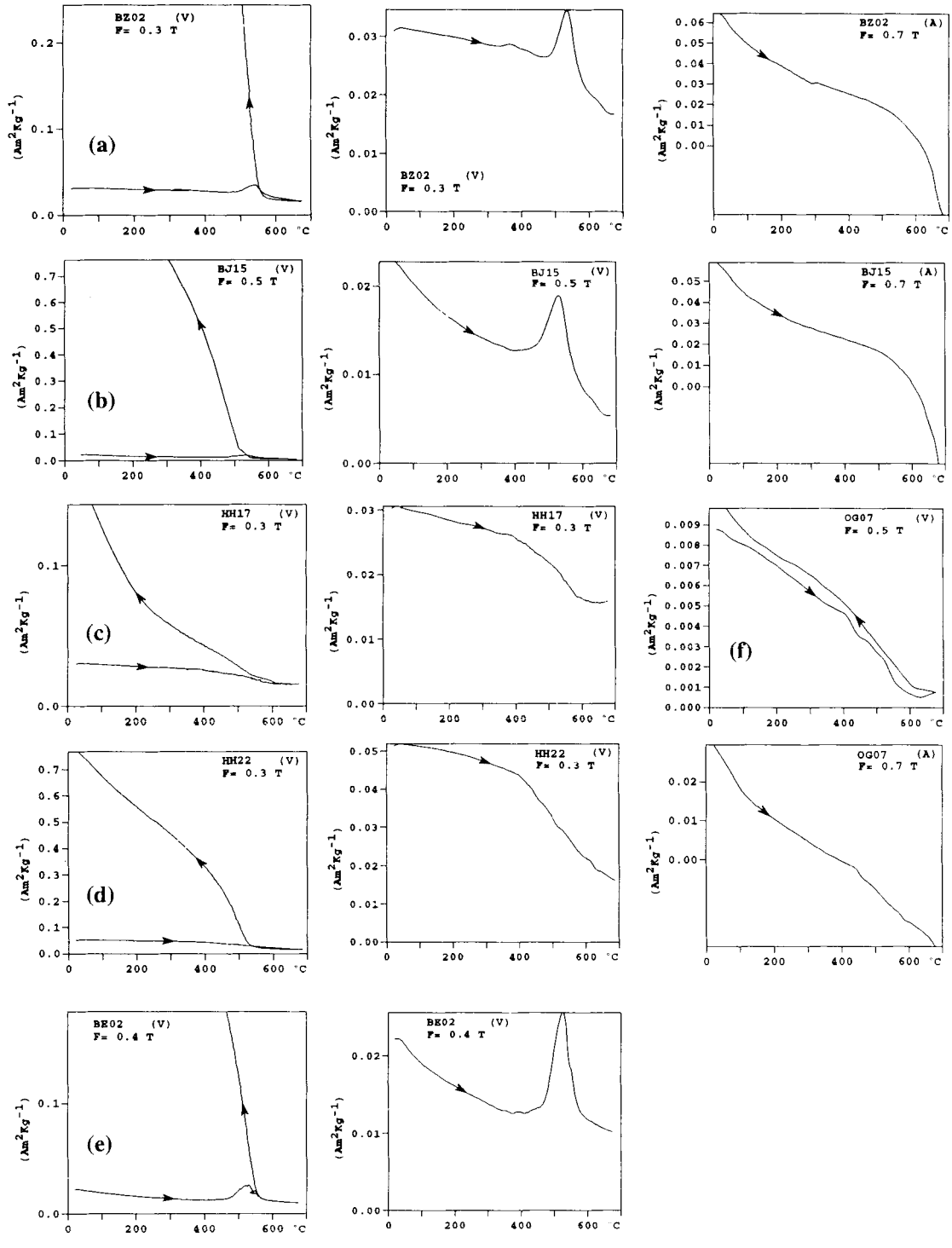


Figure 5. Thermomagnetic curves for specimens from sites OG (f) and HH (c, d) compared with those of sites BE (e), BZ (a) and BJ (b) northern the Béchar basin. (A) and (V) are, respectively, experiments at atmospheric pressure and under vacuum conditions (for explanation see text). Enlargement diagrams corresponding to the heating part of the curves (noticed V) are shown in the second column. The thermomagnetic curve of specimen OG07 (A) shows that the Curie point around 680 °C corresponds to the Curie point of haematite, which is also corroborated by the curve under vacuum conditions. The same observations are made for specimen HH22 (the heating part of the curve is only represented because weight loss occurred in free-air conditions). The heating part of the curve corresponds to the Curie point of haematite and magnetite. BZ02 (V): probably sulphides transformed into magnetite and a small amount of haematite around 500–550 °C.

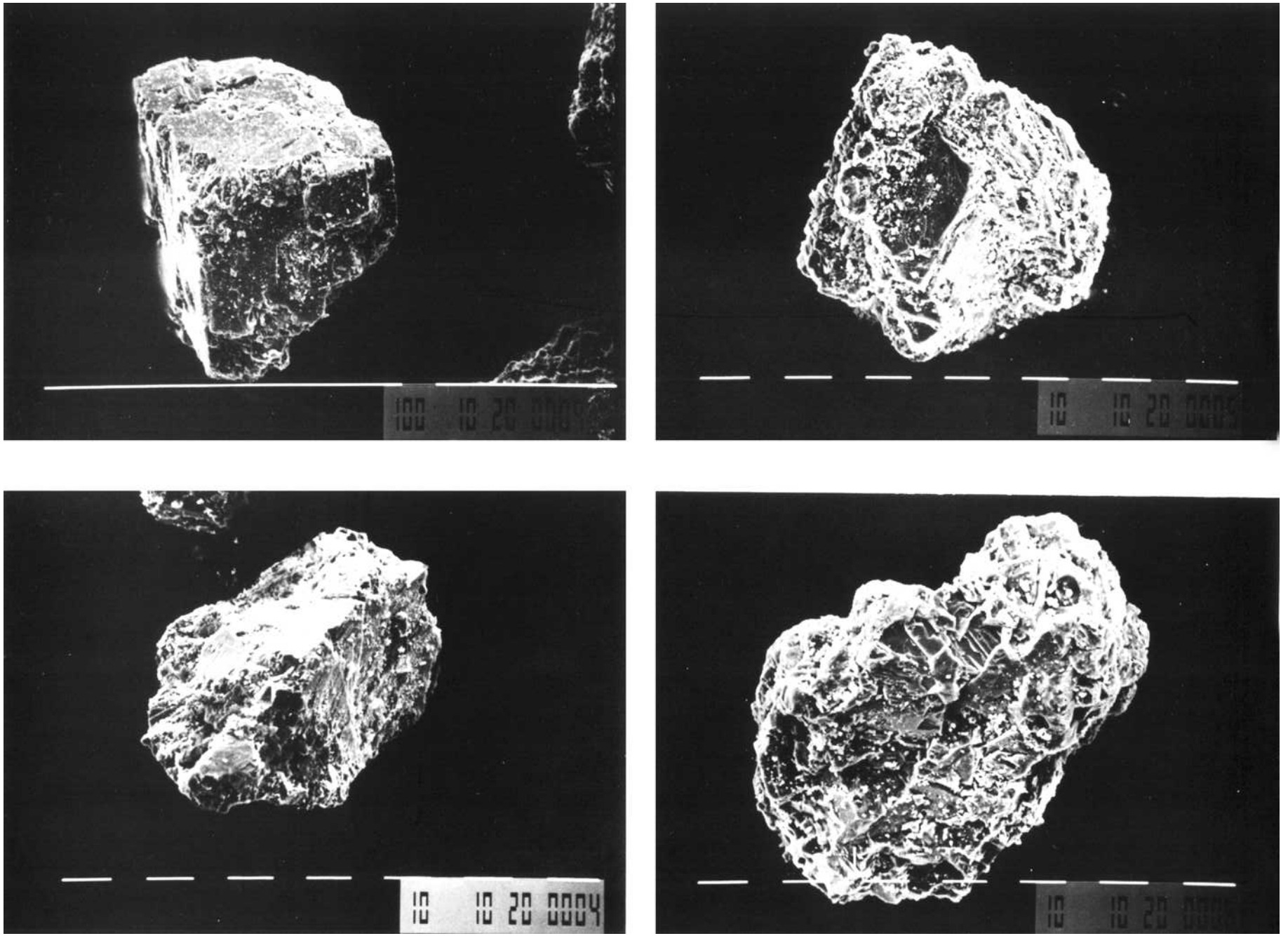


Figure 7. SEM microscopy, after magnetic separation, of a pseudomorphosed pyrite grain from site CB (a), of a magnetite grain from a specimen of site BJ (b) and haematite roses with planes from sites BJ (c) and CB (d). The scale bars are 10^{-8} m for (a) and 10^{-7} m for (b, c, d).

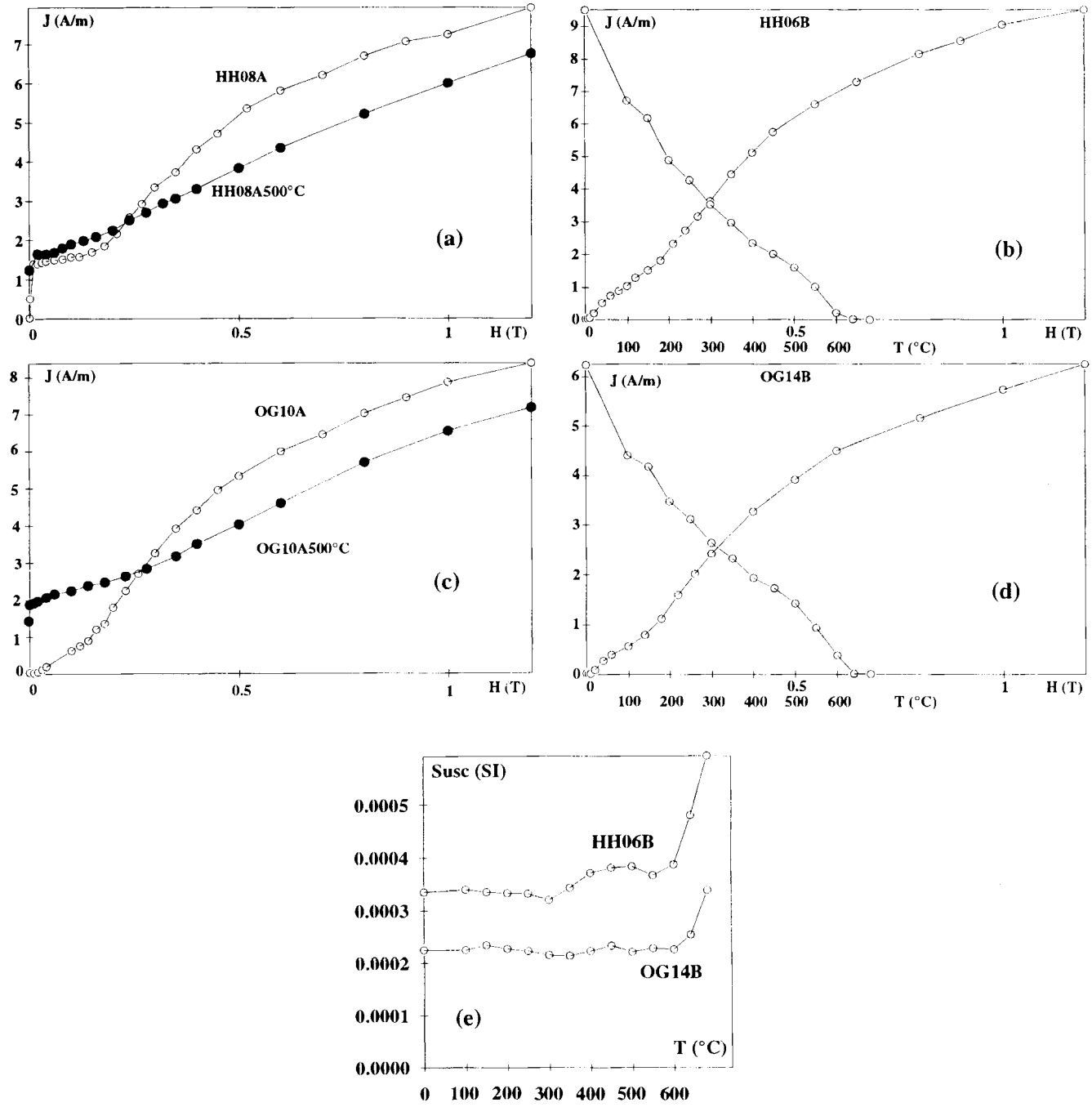


Figure 6. IRM curves versus inducing field up to 1.2 T of specimens from Ouarourout, at room temperature and at 500 °C (a, c), showing both mineralogical transformation before 500 °C (see text). Demagnetization of IRM curves (b, d) probably showing both the presence of magnetite with production of haematite suggested by mineralogic changes in susceptibility curves (e) up to 550–600 °C.

The Late Hercynian folding phase (290 Ma) affected sediments from the Béni–Abbès area only. It developed after the shortening phases, related to the collision between Laurasia and Africa in the Upper Carboniferous. This folding phase post-dated the lithospheric flexure which led to the formation of parallactic basins [e.g. coal basins of Kenadza (Algeria) and Djerada (Morocco)]. In the Lower Jurassic (Conrad 1974), an *Extensional phase* (175–190 Ma) developed. It corresponds to doleritic dike emplacement, visible in the Antar and Béchar mountains. Radiometric ages (Hailwood & Mitchell 1971; Conrad 1974) show that they are related to the first phase of Atlantic Ocean opening. This is also contemporaneous with deformations which took place along the South Atlas fault (faulting stage). The last folding phase recorded in the studied area is referred to as the *Atlasic compressional phase* (120–140 Ma). It is dated as Jurassic–Cretaceous. In the vicinity of the Béchar basin, this tectonic phase is linked to a weak tightening of the folds. The Saharan Atlas (oriented N50°) acted as an indenter along an EW direction and deformed a

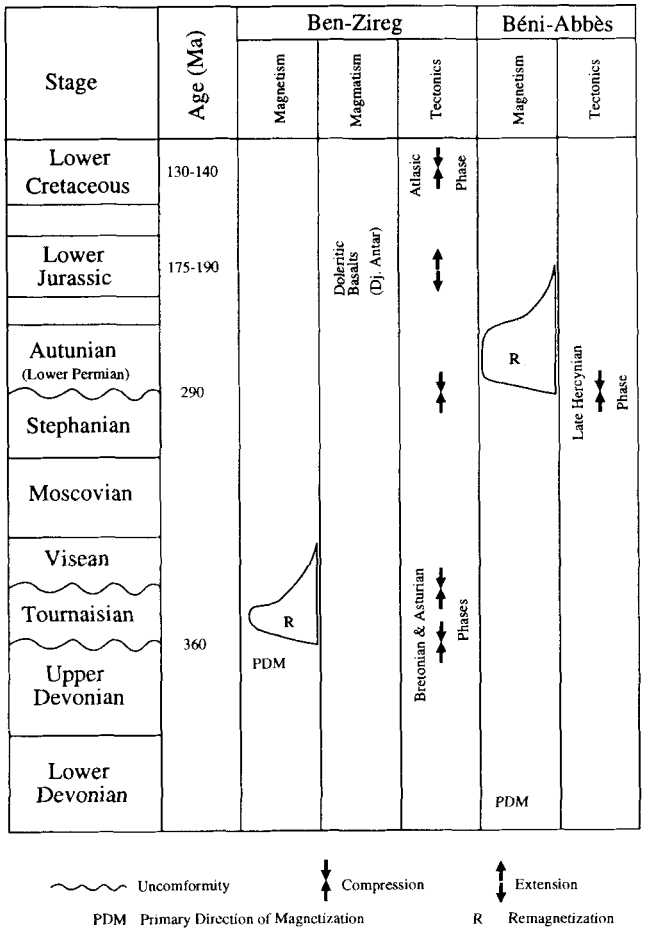


Figure 8. Tectonic phases related to rock magnetism. The geological time scale and local unconformities are taken respectively after Haq & Heysinga (1987) and Pareyn (1961). The intensity of tectonic phases are after Conrad (1974) and Conrad & Lemosquet (1984).

part of already oriented N50° folded structures. The Béni-Abbès area was not affected by this phase. In the Ben-Zireg structure, close to the Atlasic chain, only weak effect of this compressional phase can be identified, since Cainozoic and Mesozoic deposits are undeformed. To the North, the Atlasic folds become more prominent.

Perrin & Prévot (1988) proposed that the periods of overprint might not be related to any geologic or climatic phenomenon. Remagnetization mechanism may develop through less 'stable' magnetic carriers. It is common to attribute remagnetization to a wide chemical event with fluid circulation associated with major tectonic phases (here Bretonian and tardi-Hercynian orogens, Fig. 8). Reducing environment is in favour with the occurrence of changes in minerals through a complex slow mechanism. A proposed model for this region is given in Fig. 9, showing the importance of a possible low-metamorphism mechanism, which may lead to a mineralogy change. The overprint may then give a new 'hard' magnetization to the NRM, thus a noticeable high unblocking temperature (here around 500 °C), which could not be completely cleaned by a future wide overprint (e.g. during the Kiaman period).

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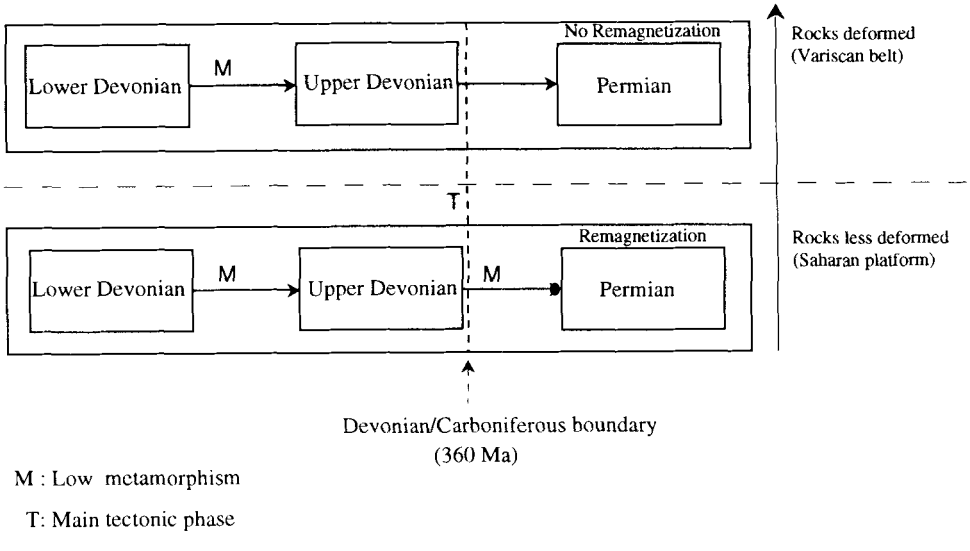


Figure 9. Possible model of space-time 'vaccination' against Permian remagnetization.

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